MIL-R-10985A (AR)
15 October 1956
SUPERSEDING
MIL-R-10985 (Ord)
22 March 1951

MILITARY SPECIFICATION

REEL, AUTOMATIC TAKE UP, EXTENSION CABLE, 24-VOLT D.C.

1. SCOPE

1.1 This specification covers spring-driven reels with extensions cable for use on military tactical vehicles and in other applications having 24-volt D.C. electrical systems. Reels are used to furnish power for lighting and other purposes at a distance from the power source. The reel provides manual pay-out and automatic take-up of the cable without interruption of power to the using device.

2. APPLICABLE DOCUMENTS

2.1 The following documents in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

NN-B-591	- Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621	- Boxes, Wood, Nailed and Lock-Corner.
QQ-A-591	- Aluminum Alloy, Die-Castings.
QQ-B-613	- Brass (Copper-Zinc Alloy), Leaded and Non-Leaded; Plate, Rolled Bar, Sheet and Strip
QQ-B-626	- Brass, Leaded and Non-Leaded (Copper Zinc Alloy) Rods, Bars, Shapes, and Forgings.
QQ-C-533	- Copper-Beryllium Alloy Strip.
QQ-M-151	- Metals; General Specification for Inspection of.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
QQ-S-365	- Silver Plating (Electrodeposited).
QQ-S-00640	- Steel, Carbon; Sheet and Strip.
TT-C-595	- Colors; (for) Ready-Mixed Paints.
TT-E-485	- Enamel; Semi-Gloss, Rust-Inhibiting.
TT-E-489	- Enamel; Gloss, Synthetic (for Exterior and Interior Surfaces).
TT-E-529	- Enamel, Synthetic, Semi-Gloss.
TT-P-636	- Primer Coating, Synthetic, Wood and Ferrous Metal.
TT-P-666	- Paint; Primer, Zinc Yellow, for Aluminum and Magnesium Surfaces.
WW-T-791	- Tubing, Brass, Seamless.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

FSC 6150

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Military

JAN-P-106	- Boxes; Wood, Nailed, (Overseas Type).
MIL-B-107	- Boxes, Wood, Wirebound (Overseas Type).
MIL-P-106	- Preservation, Methods of.
JAN-P-127	- Packaging and packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-B-138	- Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for Weight of Contents not Exceeding 500 Pounds).
MIL-S-901	- Shockproof-Equipment, Class HI (High-Impact); Shipload
MIL-V-3329	- Vehicles, Combat,Self-Propelled and Towed; General Requirements for.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-11415	- Steel Parts; Ordnance.
MIL-C-13486	- Cable, Vehicle: Low Tension, 30-Volt, Heavy-Duty, Single Conductor and Multiconductor.
MIL-E-13856	- Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Specifications for.
MIL-F-13927	- Fungus Resistance Test; Automotive Electrical Components, General Specification for.
MIL-C-13993	- Connector, Plug, Electrical; Connector, Receptacle, Electrical (Waterproof-for Military Vehicles).
MIL-P-15037	- Plastic-Material, Laminated Thermosetting: Sheets, Glass-Cloth Melamine-Resin.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

Ordnance Corps

D7262199	- Extension Ass'y, Reel.

(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction samples. - Three preproduction samples of each reel proposed to be furnished shall be forwarded to the place spcified by the procuring activity for inspection and testing to assure conformance to the requirements specified herein. After approval no changes shall be made unless authorized by the procuring activity.

3.2 Materials.- The materials shall be as spcefied herein, on the applicable drawings, and in the applicable specifications. Materials not specifically covered by this or applicable specifications shall be capable of use at any ambient air temperature within the range of minus 65°F. to plus 160°F. Materials which may warp or absorb moisture shall not be used.

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3.3 Design. - The reel shall conform to Drawing D7262199 (see 2.1).

3.4 Performance. - The reel shall operate when mounted in any position.

3.4.1 Extreme temperature operation. - Reels shall meet the requirements throughout an ambient air temperature range of minus 65°F to plus 160°F., when tested as specified in 4.5.1.

3.4.2 Resistance to effects of water. - Reel shall conform to type I, class 4 components of Specification MIL-E-13856 and shall resist effects of water and temperature cycling when tested as specified in 4.5.9. Reel shall operate as specified and shall show no insulation breakdown after testing.

3.4.3 Take-up velocity. - The take-up velocity of the reel shall be not less than 150 f.p.m., as determined by the test specified in 4.5.3.

3.4.4 Effect of wear. - The reel shall show no evidence of damage or excessive wear and shall maintain electrical continuity, and the voltage drop across both slip rings and brushes shall be not more than 400 millivolts after testing as specified in 4.5.4.

3.4.5 Effect of vibration. - The reel shall withstand the test specified in 4.5.5. without showing evidence of failure of component materials or assemblies. During the test there shall be no interruption of current as evidenced by flickering of the lamp.

3.4.6 Effect of Shock. - The reel shall withstand the test specified in 4.5.6 without showing evidence of failure of component materials and assemblies. Subsequent to the test there shall be no interruption of current as evidenced by flickering of the lamp.

3.4.7 Resistance to corrosion. - When tested as specified in 4.5.7, the reels shall meet the requirements of 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5, 3.4.6, 3.4.8, and 3.5.6.

3.4.8 Resistance to fungi. - When tested as specified in 4.5.8, the reels shall meet the requirements of 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5, 3.4.6, 3.4.7, and 3.5.6.

3.5 Components. - The reel shall consist of the following components: housing, cable drum, cable, receptacles, clockspring motor, current collector, brake mechanism, and fairlead.

3.5.1 Housing. - The housing, of two-piece construction, shall be fabricated of sheet steel to the contour and dimensions shown on the applicable drawing. Two sheet steel straps shall be fastened to the bottom of the housing to provide means for attaching the reel in its intended location. Sheet steel used in fabrication of the housing and straps shall be not less than 0.0478 inch thick and shall conform to Specification MIL-S-11415.

3.5.2 Cable drum. - The drum shall be of two-piece construction. Capacity of the drum shall be 25 feet of cable.

3.5.3 Cable. The cable provided with the reel shall be of the size, length, and type shown on the applicable drawing. Cable shall conform to MIL-C-13486. One end of the cable shall be connected to the brushes on the drum and the other end to the output connector or terminals (see 3.5.4). A snubber shall be provided at the free end of the cable, 3 inches from the plug, to prevent cable tension from affecting terminals of the plug in retraction of the cable. Cable used in connecting the slip rings (see 3.5.6.1) to the input receptacle shall conform to Specification MIL-C-13486.

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3.5.4 Electrical connections. - The reel shall be provided with an electrical input receptacle and output connector or terminals, as specified (see 6.1), of the type shown on the applicable drawing. Receptacles and connectors shall conform to specification MIL-C-13993. The receptacle and terminals, or connectors, shall be attached to the reel and cable as specified on the applicable drawing.

3.5.5 Motor. - The reel motor shall be a clockspring of the heavy-duty type, mounted in a sealed compartment underneath the drum in such a manner as to provide a balanced flywheel. The spring shall maintain cable tension at not less than one pound when the cable is fully retracted and not less than five pounds when the cable is fully extended. Spring shall be fabricated of steel conforming to composition FS 1065, FS 1075, or FS 1095 of Specification QQ-S-00640 and shall be heat-treated to a hardness of Rockwell A71 to 75 or equivalent.

3.5.5.1 Lubrication. - Spring shall be lubricated for life with a grease conforming to Specification MIL-G-10924.

3.5.6 Current collector. - The current collector, consisting of slip rings and brushes, shall be mounted in a sealed compartment underneath the drum to maintain electrical continuity during static or dynamic operation of the reel when tested as specified in 4.5.2.

3.5.6.1 Slip rings. - The two slip rings shall be fabricated of brass of composition conforming to Specification QQ-B-626, QQ-B-613 or WW-T-791. Rings shall be of such construction as to assure against their becoming eccentric in service. The rings shall be concentric, have a smooth surface free from hard spots, and shall be silver plated to a thickness of not less than 0.005 inch in accordance with type II of Specification QQ-S-365. Rings shall be insulated from each other and from the housing, and shall be securely attached to the housing.

3.5.6.2 Brushes. - Two brushes of the dual type shall be attached to a terminal block on the bottom of the drum. Brushes shall be fabricated of beryllium-copper conforming to condition HT of Specification QQ-C-533. Contact points shall be silver, spun attached to the brushes. Terminal block shall be plastic conforming to Specification MIL-P-15037.

3.5.7 Brake mechanism. - The reel shall be provided with a ratchet stop so that the cable may be extended and set at any desired length to hold itself against the tension of the reel. The ratchet stop shall be arranged to lock the cable at least twice with every revolution of the drum. A pull of not greater than 7-1/2 pounds shall release the pawl of the stop either for automatic retraction or further extension of the cable. Pawl shall be fabricated of brass of a composition and temper that will withstand forces applied against it by the ratchet stop.

3.5.8 Fairlead. - The reel shall be provided with an automatic fairlead geared to the drum, to assure even winding of the cable on the drum. The fairlead shall be installed in a support attached to the housing. The support shall be an aluminum allloy die casting of composition 5, 10, 11, or 12 conforming to Specification QQ-A-591.

3.6 Dimensions. - Over-all dimensions of the reel shall be as shown on the applicable drawing.

3.7 Finishes.

3.7.1 Painting.

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3.7.1.1 Prime Coat.- All exterior metal surfaces of the reel shall be given one coat of primer having a film thickness of 1.0 to 1.5 mils. For ferrous-metal parts the primer shall conform to Specification TT-P-636. For aluminum or aluminum-alloy parts the primer shall conform to Specification TT-P-666.

3.7.1.2 Finish coat. - Finsih coat shall conform to spot 2430 of Specification TT-C-595 using a semi-gloss enamel confoming to Specification TT-E-485 or TT-E-529, or spot 1755 of Specification TT-C-595 using a gloss enamel confoming to Specification TT-E-489, as specified (see 6.1).

3.7.2 Chemical or metal finishes. - All brass and steel parts not otherwise finished shall be cadmium plated in accordance with type II, class A or B of Specification QQ-P416.

3.8 Marking. - Each reel shall be marked with the following information stamped on the outside of the housing:

Ordnance part number.
Manufacturer's name or trade-mark.
Manufacturer's part number.
Specification number.
Date of manufacture.

3.9 Workmanship. - All machined surfaces shall be smoothly finished and be free from sharp edges or burrs. Particular attention shall be paid to neatness and thoroughness of soldering connecting wires and other conductors, marking of parts, plating, painting riveting, tightness of threaded and other fastening devices, and ruggedness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction sample. - The preproduction sample (see 3.1) shall be subjected to the follwoing tests in the order listed:

Specimen	Test
A	4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.6, 4.5.9.
B	4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.6, 4.5.7, 4.5.9.
C	4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.5.5, 4.5.6, 4.5.8, 4.5.9.

4.1.1 Rejection. - Failure of a preproduction sample to pass all specified tests shall be cause for refusal by the Government to conduct further tests until it has been proved to the satisfaction of the procurring activity that faults revealed by the tests have been corrected.

4.2 Sampling.

4.2.1 Lot size. - A lot shall consist of all reels of one part number from an identifiable production period, from one manufacturer, and submitted at one time for acceptance.

4.2.2 Inspection. - A sample shall be selected from each lot in accordance with inspection level II of Standard MIL-STD-105 under the single sampling plan.

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4.2.3 For acceptance test. - Sampling for acceptance tests (see 4.4.1) shall be in accordance with inspection level L6 of the appendix to Standard MIL-STD-105 under the single sampling plan.

4.3 Inspection.

4.3.1 Visual, dimensional, and primary functional inspection. - Each reel selected in accordance with 4.2.2 shall be inspected for conformance to the following acceptable quality levels (A.Q.L.) on the basis of percent defective:

Classification	A.Q.L.
Major	1.0
Minor	2.5

4.3.1.1 Classification of defects. - For inspection pruposes defects shall be classified as follows:

Major defects (1.0 percent AQL)
- Take-up velocity (see 3.4.3)
- Brake mechanism (see 3.5.7)
- Current collector continuity (see 3.5.6)

Minor defects (2.5 percent AQL)
- Overall dimensions (see 3.6)
- Finsih (see 3.7)
- Marking (see 3.8)
- Workmanship (see 3.9)

4.4 Tests.

4.4.1 Acceptance tests. - Samples selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.5.2 and 4.5.3. The acceptable quality level used for basis of rejection shall be 6.5 (based on percent defective) under the normal inspection plan specified in Standard MIL-STD-105.

4.5 Test procedures.

4.5.1 Extreme temperature tests.

4.5.1.1 High Temperature.- The reels hall be placed in a chamber and heated for such period of time as is required for the temperature of the reel to become stabilized at $160^0 \pm 2^0$F. Subsequent to the heating period and while still held at a temperature of $160^0 \pm 2^0$F., the reel shall be subjected to test specified in 4.5.3 to determine compliance with 3.4.1.

4.5.1.2 Low temperature. - The reel shall be subjected to the test specified in 4.5.1.1 after the reel has been stabilized for 24 hours at a temperature of minus $65^0 \pm 2^0$F. to determine compliance to 3.4.1.

4.5.2 Contnuity test.- Terminals or contacts at output end of cable shall be shorted together and an ohmmenter shall be connected across input receptacle contacts. The cable shall be withdrawn and allowed to rewind 2 times and a continuity check shall be made during these actions to determine conformance to 3.5.6.

4.5.3 Take-up velocity test. - With the reels attached to a firm support, the cable shall be withdrawn to its most extended length and allowed to rewind. The time required for the reel to automatically take up the the cable shall be measured to determine compliance with 3.4.3.

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4.5.4 Wear tests.

4.5.4.1 - Apparatus. - The apparatus shall consist of an approved device arranged to withdraw not less than four feet of cable from the reel and allow it to rewind. A lamp load of 10 amperes and a 28.5-volt, d.c. power source shall be attached to the reel during the testing.

4.5.4.2 - Procedure. - With the reel attached to a firm support and with load applied, the device shall be operated in such a manner as to draw approximately four feet of cable from the reel and allow it to be reround. This shall constitute one cycle of operation and shall be repeated 50,000 times. Subsequent to such operation, the reel shall be tested by the method specified in 4.5.2 to determine conformance to 3.4.4. If cable fails before completion of the cycling a new segment of cable shall be used to complete the testing. Failure of the cable alone shall not be construed as failure of the reel itself.

4.5.5 Vibration test.

4.5.5.1 Apparatus. - The apparatus shall consist of an approved device for mounting the reel and vibrating it through a total maximum amplitude of 0.060-inch at 10 to 55 cycles per second (c.p.s.).

4.5.5.2 Procedure. With the reel connected to a 28.5 volt, d.c., electrical supply and a lamp load connected to the cable plug, the reel shall be rigidly mounted on the vibrating device with the lamp permitted to hang free. The reel shall then be subjected to vibrations with a miximum amplitude of 0.030-inch (0.060-inch total travel) through a frequency range of 10 to 55 c.p.s. The reel shall be vibrated for a period of 1 hour in the direction of each of the three major axes with continually varying frequency, with the entire range of frequencies being transversed each minute. During the test the cable shall be partially unwound and retracted, and the lamp observed for evidence of flickering to determine conformance to 3.4.5.

4.5.6 Shock test. - The reel shall be subjected to the shock test for type A, lightweight equipment specified in Specification MIL-S-901. Subsequent to the test, the reel shall be connected to a 28.5 volt, D.C. electrical supply and a lamp load connected to the cable plug. The lamp shall be observed for evidence of flickering to determine conformance to 3.4.6.

4.5.7 Corrosion-resistance test. - The reel shall be subjected to the corrosion-resistance test specified in Specification QQ-M-151 for a period of 200 hours. At the conclusion of the 200-hour period, the reel shall be opened and inspected for evidence of corrosion to determine conformance to 3.4.7.

4.5.8 Fungus-resistance test. - The reel shall be subjected to the fungus-resistance test specified for class 2, method B in Specification MIL-F-13927. Subsequent to the test, the reel shall be opened and inspected for evidence of the growth of fungi to determine compliance with 3.4.8.

4.5.9 Resistance to water effects test. - Reel shall be subjected to the applicable tests specified for type I, class 4 in Specification MIL-E-13856, and shall be operated under a 10-ampere lamp load for 30 minutes when electrical operation is indicated, to determine conformance to 3.4.2.

5. PREPARATION FOR DELIVERY

5.1 Level of preservation and packaging, and level of packing. - Cable reels shall be prepared for delivery in accordance with the level of preservation and packaging, and the level of packing specified (see 6.1).

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5.2 Preservation and packaging.

5.2.1 Level A and level C.

5.2.1.1 Cleaning. - The reels shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.2.1.2 Drying. - Drying shall be accomplished by any applicable method specified in Specification MIL-P-116.

5.2.1.3 Packaging.- All exposed connectors shall be sealed with a tape conforming to Specification JAN-P-127 or with plastic caps which will afford the same degree of protection. The reels shall then be packaged in accordance with method III of Specification MIL-P-116.

5.3 Packing.

5.3.1 Level A.- Reels shall be packed in containers conforming to Specification PPP-B-601 (overseas type), JAN-P-106, MIL-B-107, or MIL-B-138. The gross weight shall not exceed 150 pounds.

5.3.2 Level B. - Reels shall be packed in containers conforming to Specification PPP-B-601 (domestic type), NN-B-591, NN-B-621, or PPP-B-585. The gross weight shall not exceed 150 pounds.

5.3.3 Level C - Reels shall be packed in containers that will assure mechanical protection and be acceptable to common carrier for safe delivery.

5.4 Marking. - In addition to any special marking required (see 6.1), interior packages and shipping containers shall be marked in accordance with applicable requirements of Standard MIL-STD-129.

5.4.1 Level A. - All exterior marking shall be coated with a transparent-waterproof compound as specified in Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Number of preproduction samples, if more than three (see 3.1).
c. Output connector or terminals (see 3.5.4).
d. Color required (see 3.7.1.2).
e. Level of preservation and packaging, and level of packing required (see 5.1).
f. Special markings, if required (see 5.4).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-10985A Reel, Automatic Take-Up, Extension Cable 24-Volt D.C.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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